Gesture Controlled Wheelchair

Group 02

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INTRODUCTION

The Gesture-Controlled Wheelchair project introduces an innovative and assistive technology solution that empowers individuals with mobility impairments to navigate their environment using intuitive hand gestures. By leveraging advanced gesture recognition technology, the system enables users to control their wheelchairs with precision and ease, offering a more natural and intuitive alternative to traditional joystick control methods [1].

OBJECTIVES

The objective of the Gesture Controlled Wheelchair project is to develop an intuitive and accessible assistive technology solution that enables individuals with mobility impairments to control their wheelchairs using hand gestures [2].



Fig 1. Objectives of gesture controlled wheelchair.

Additional key objectives include:

1. Enhanced Independence
2. Enhanced mobility & Gesture-Based Control
3. Increased convenience & comfort
4. Improved quality of life

SCHEMATIC DIAGRAMS

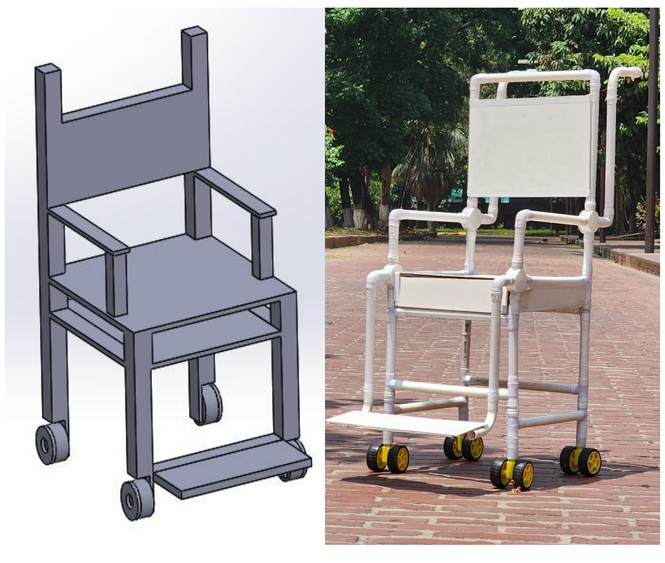


Fig 2. Schematic Diagram of cad and real model of the gesture controlled wheel chair.

METHODOLOGY

The Wheelchair circuit was carefully put together to guarantee correct operations and dependability. The wheelchair frame has the Arduino Uno, the primary control unit, fixed firmly in place. The wheelchair can easily be controlled remotely thanks to the precise connection between Arduino Uno and the Bluetooth

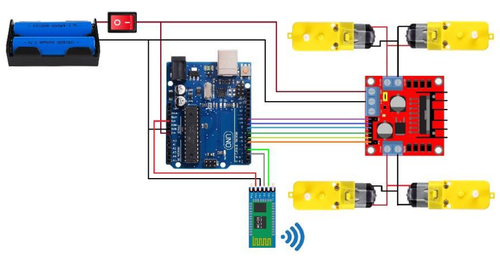


Fig 3. Schematic Diagram of wheelchair circuit

The wheelchair's motors and Arduino Uno were precisely wired to accept the motor driver L293DN, which is essential for motor control, into the circuit.In order to control the power supply and allow the wheelchair to operate safely, battery switches were integrated into the circuit.

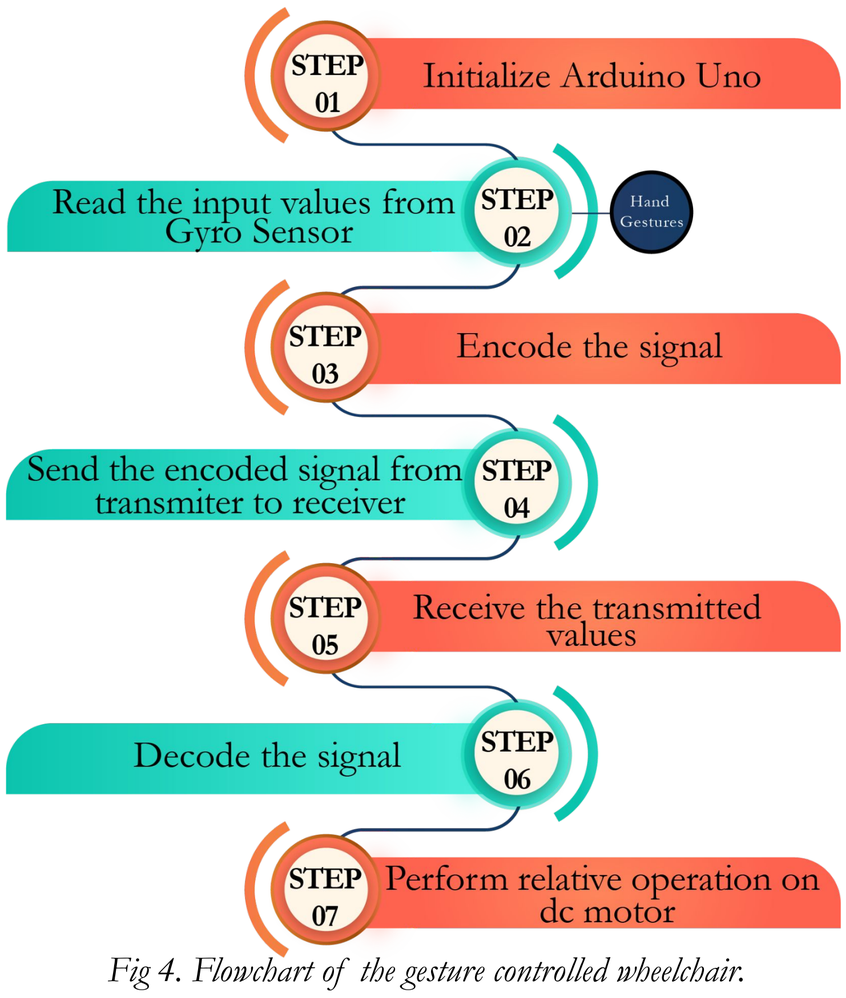


Fig 4. Flowchart of the gesture controlled wheelchair.

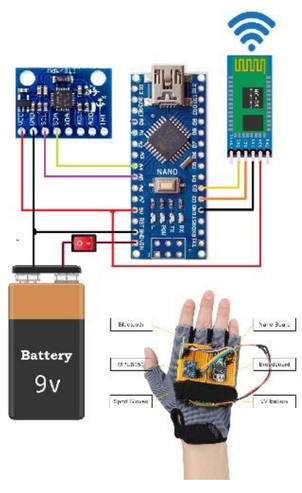


Fig 5. Schematic Diagram of the hand's circuit.

To maximize space and enable effective wiring for hand's circuit , the Arduino Nano, battery, MPU650 accelerometer gyroscope, and Bluetooth module—were arranged on the breadboard in an organized way.Every component was positioned with diligence to minimize interference and provide the best possible signal transmission and reception. Carefully soldered or fastened to the breadboard, the wiring connections were made to avoid loose connections and signal deterioration.

Appropriate insulating methods were used to protect against electrical dangers and short circuits. The MPU650 accelerometer gyroscope gesture data was read and interpreted using Arduino code. Signal transmission between the gesture-controlled wheelchair system and the hand gesture control circuit was made possible by the use of Bluetooth communication [1][2].

RESULTS & DISCUSSION

1. The gesture system accurately interpreted hand gestures, allowing to control wheelchair movements with precision
2. The chair moved smoothly on the surface in response to command given by gesture.
3. Straight line and turning speed has been kept constant at a manageable level..
4. Some small tuning was done to the chair after final assembly.



Fig 6. Forward motion



Fig 7. Backward motion



Fig 8. Left turn

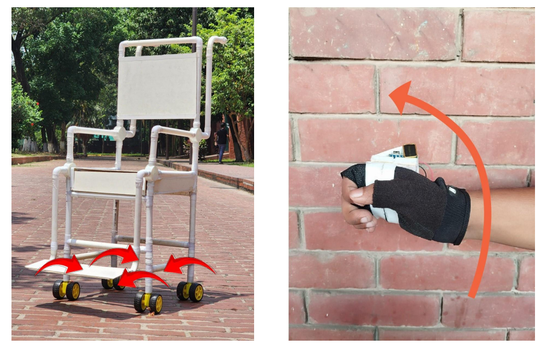


Fig 9. Right turn

Future scope:
Speed controller can be added to get different modes of speed. Camera can be added for full autonomous navigation and laser scanner can be added to avoid collision. Robotic arm can be installed on the wheelchair for more assistance to the user.[2]

REFERENCES

1. Dr. C.K. Gomathy, "The Gesture Controlled Robot", IRJET, Vol:08, Issue: 04, Apr 2021
2. Anilkumar, R. (2023). HAND GESTURE CONTROLLED WHEELCHAIR. Issue: 24.07.2023